THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 9–12: **Methods of Separation in Everyday Life, Living Creatures: Exploring their Characteristics, Nature's Treasures, Beyond Earth** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** magnet	Removes iron.
2	**(c)** it sprouts when given water	Germination indicates life.
3	**(d)** forests	Forests can regrow but not if cut faster than they grow.
4	**(b)** Mercury, Venus, Earth, Mars	Order.
5	**(c)** unsaturated	More solute can dissolve.
6	**(c)** pupa	Metamorphosis.
7	**(d)** glass bottles	Bottles can be used again.
8	**(b)** complete one revolution around the Sun	Orbital period.
9	**(d)** to avoid spurting	Splashing.
10	**(d)** reproduction	Life continues.
11	**(d)** 21 L	About 21 per cent.
12	**(c)** it rotates once in the time it orbits the Earth	Synchronised.
13	**(a)** residue	It stays on paper.
14	**(b)** seed is living	Growth.
15	**(c)** oxygen	Combustion needs oxygen.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Sand does not dissolve (1); Sedimentation and filtration are suitable (1).	1+1
17	It germinates under suitable conditions (1); Its life processes are slow (1).	1+1
18	It contains many organisms (1); It supports plant growth (1).	1+1
19	Day is one rotation (1); Year is one revolution (1).	1+1
20	Both dissolve (1); They pass through together (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Handpick stones (1); Use magnet for nails (1); Sieve out sand (1).	1+1+1
22	Fire does not have cells (1); It does not reproduce on its own (1); It cannot respond or repair itself (1).	1+1+1
23	Light a candle (1); Cover with a jar (1); It goes out after some time (1).	1+1+1
24	Earth has air and water (1); Moon has none (1); Moon has extreme temperatures (1).	1+1+1
	OR: Large surface area (1); Sun heats quickly (1); Faster evaporation (1).	1+1+1
25	Plants make food in leaves (1); Animals cannot (1); They depend on plants (1).	1+1+1
	OR: Do not decay (1); Block drains (1); Harm animals (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Magnet removes iron (1); Add water (1); Sawdust floats and is skimmed (1); Filter sand (1); Evaporate to get sugar (1).	1+1+1+1+1
	OR: Place a potted plant near a window (1); Rotate it after some days (1); It bends towards light (1); Rotate again (1); It bends again (1).	1+1+1+1+1
27	Take equal volumes in funnels (1); Pour equal water (1); Measure water collected (1); Clay holds the most (1); Sand holds the least (1).	1+1+1+1+1

	OR: Locate Saptarishi (1); Extend line from the bowl (1); Pole star gives north (1); New moon starts month (1); Full moon near mid-month (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 9–12: **Methods of Separation in Everyday Life, Living Creatures: Exploring their Characteristics, Nature's Treasures, Beyond Earth** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** Venus: hottest planet	Right.
2	**(a)** crystallises out	Solubility falls.
3	**(b)** water	Aquatic eggs.
4	**(d)** aluminium cans	Metal recycled.
5	**(d)** rotate once on its axis	Rotation period.
6	**(d)** to hold the paper well	Good contact.
7	**(b)** cells	Cellular.
8	**(d)** 156 L	78 per cent of 200.
9	**(b)** 29.5 days	Phase cycle.
10	**(c)** salt solution	Salt dissolved.
11	**(c)** no, it cannot grow, respire or reproduce	Machines lack life processes.
12	**(b)** nitrogen	Inert.
13	**(c)** its temperature is suitable	Habitable zone.
14	**(d)** solution became saturated	Extra salt remains.
15	**(b)** gills	Larvae have gills.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	No light for food making (1); Chlorophyll fails to form (1).	1+1
17	Reduce: use less (1); Reuse: use again (1).	1+1
18	They are in free fall around Earth (1); Gravity still acts (1).	1+1
19	It remains behind (1); The clear liquid is the filtrate (1).	1+1
20	Gills work in water (1); It cannot take oxygen from air (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	It orbits Earth (1); It was not made by humans (1); Artificial ones are man-made (1).	1+1+1
22	Dyes differ (1); Colours stay mixed (1); Special methods needed (1).	1+1+1
23	Body gets hot (1); Sweat removes waste and heat (1); Helps cooling (1).	1+1+1
24	Solar (1); Wind (1); Biogas (1).	1+1+1
	OR: Collects light (1); Focuses it (1); Magnifies image (1).	1+1+1
25	Heat a few drops on a glass slide (1); Water evaporates (1); White solid remains (1).	1+1+1
	OR: (a) Germinate and grow (1); (b) No life process (1); (c) Grows and reproduces by spores (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Winnowing and sieving (1); Straining (1); Evaporation (1); Churning (1); Basic ideas same (1).	1+1+1+1+1
	OR: Plants: stomata (1); Fish: gills (1); Humans: lungs (1); All take in oxygen (1); Release carbon dioxide (1).	1+1+1+1+1
27	Population and vehicles rise (1); Fossil fuels deplete (1); Use solar (1); Use efficient devices (1); Save energy (1).	1+1+1+1+1
	OR: Water in liquid form (1); Oxygen (1); Right temperature (1); Protection by atmosphere (1); Distance from the Sun (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 9–12: **Methods of Separation in Everyday Life, Living Creatures: Exploring their Characteristics, Nature's Treasures, Beyond Earth** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** it saves ore and energy	Saves resources.
2	**(a)** Neptune	Farthest from the Sun.
3	**(d)** pure water is collected	Distillation.
4	**(a)** cell, tissue, organ, organism	Standard order.
5	**(a)** being depleted	Overuse.
6	**(b)** 15 January	About two weeks later.
7	**(b)** evaporation	Heat.
8	**(a)** it can germinate in suitable conditions	Dormancy.
9	**(d)** oxygen is used up	Needs oxygen.
10	**(b)** sound cannot travel	No air.
11	**(b)** evaporation of salt solution	Salt can be recovered.
12	**(b)** lungs and skin	Adaptation.
13	**(a)** using public transport	Fewer vehicles.
14	**(d)** Mercury	Nearest to the Sun.
15	**(c)** centrifuge	Spins.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	They do not run out (1); They cause less pollution (1).	1+1
17	They are closer (1); They look like discs (1).	1+1
18	Use a magnet to remove iron (1); Sugar remains (1).	1+1
19	It does not grow or reproduce (1); It is made, not born (1).	1+1
20	Roots bind the soil (1); Trees give oxygen (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Heat a lump of soil in a test tube (1); Droplets of water form (1); Bubbles appear when soil is put in water (1).	1+1+1
22	Moon orbits Earth in about a month (1); Lit part seen changes (1); Cycle repeats (1).	1+1+1
23	Pure substance has one kind of matter (1); Mixture has more (1); Examples: gold vs sea water (1).	1+1+1
24	Sunflower follows the Sun (1); Flowers open and close (1); Roots grow towards water (1).	1+1+1
	OR: Air (1); Water (1); Come without human making (1).	1+1+1
25	Moon's gravity is weaker (1); About one sixth of Earth's (1); Mass does not change (1).	1+1+1
	OR: If solid is insoluble: filtration (1); If dissolved: evaporation (1); Check solubility (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Take two glasses with equal water (1); One cold, one warm (1); Add salt spoon by spoon (1); Count spoons dissolved (1); Warm water dissolves more (1).	1+1+1+1+1
	OR: All grow and reproduce (1); Plant makes food (1); Animal eats others (1); Mushroom absorbs from dead matter (1); Only animals move around (1).	1+1+1+1+1
27	Deforestation (1); Heavy rains (1); Loss of fertile soil (1); Terrace farming (1); Afforestation (1).	1+1+1+1+1
	OR: Earth: life, water (1); Moon: no air (1); Mars: thin air, cold (1); Venus: hot (1); Jupiter: gas, no surface (1).	1+1+1+1+1

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